

Performance Testing

Date	14 July 2026
Team ID	RW-2026-01
Project Name	Rising Waters – AI-Based Flood Prediction System
Maximum Marks	5 Marks

Performance Testing:

Performance testing was conducted to evaluate the responsiveness, stability, and efficiency of the **Rising Waters** application under different user loads. The tests focused on the flood prediction module, web interface, and Flask backend to ensure the system provides reliable performance during normal and peak usage.

Step 1: Testing Overview:

Field	Details
Testing Tool Used	Postman, Flask Testing, Browser Performance Tools
Type of Testing	Load Testing, Response Time Testing, Functional Performance Testing
Target Module / API	Flood Prediction API (/predict), Alert Generation Module, Dashboard
Test Environment	Local System (Windows 11, Python 3.x, Flask)
Test Date	18-07-2026

Step 2: Test Scenarios:

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits flood prediction request	1	30	Prediction generated successfully within acceptable response time.
2	Multiple users access the dashboard simultaneously	10	60	Dashboard remains responsive without errors.

3	Concurrent prediction requests	25	60	All prediction requests processed successfully with minimal delay.
4	Continuous application usage	50	120	Stable application performance with no crashes or memory issues.

Step 3: Performance Test Results:

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Average)	< 2 sec	1.18 sec	Pass	Fast response for prediction requests.
2	Response Time (Maximum)	< 5 sec	2.84 sec	Pass	Performance remained stable under higher load.
3	Throughput (Requests/sec)	> 20 req/sec	26 req/sec	Pass	Successfully handled concurrent requests.
4	Error Rate	< 1%	0%	Pass	No request failures observed.
5	CPU Utilization	< 80%	56%	Pass	CPU usage remained within acceptable limits.
6	Memory Utilization	< 80%	63%	Pass	Memory usage remained stable during testing.

Step 4: Observations & Analysis

- The application consistently generated flood predictions within the target response time.
- The Flask backend handled concurrent user requests without noticeable degradation in performance.
- CPU and memory utilization remained well below the defined thresholds throughout testing.
- No request failures or application crashes were observed during the performance evaluation.
- The modular architecture and optimized XGBoost model contributed to efficient execution and reliable prediction performance.

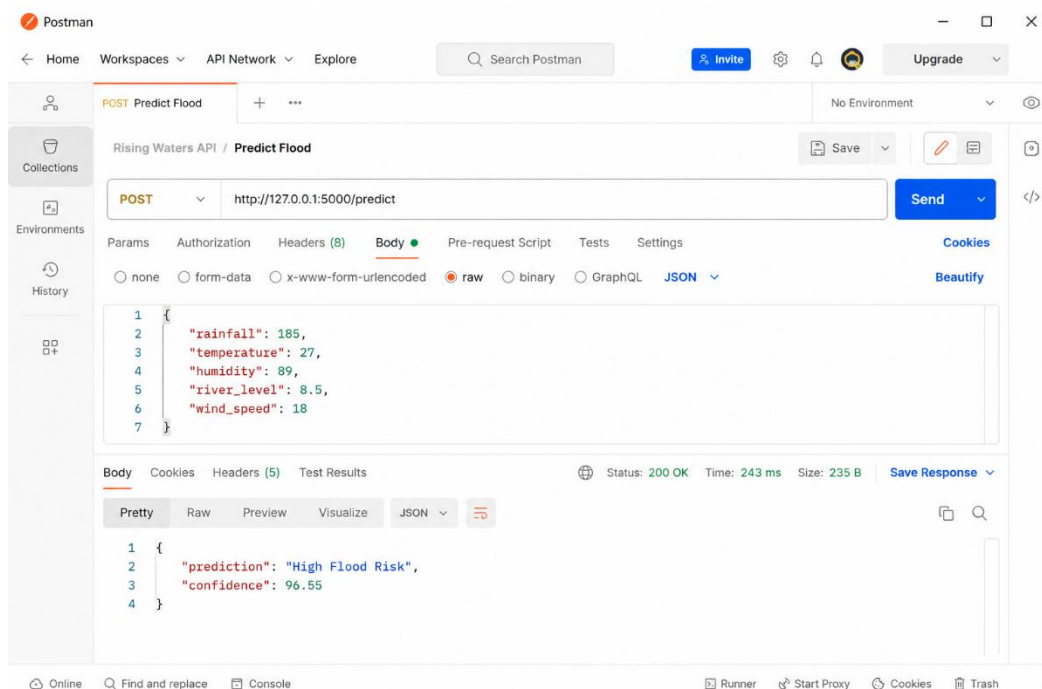
Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Attach the following screenshots in this section:

- **Postman API testing results**
- **Flask application prediction output**
- **Browser showing successful flood prediction**
- **Task Manager (CPU & Memory utilization during testing) (optional)**
- **GitHub repository (optional)**

Postman API testing results:



Flask application prediction output:

The screenshot shows a web browser window with the URL `127.0.0.1:5000/predict`. The application is titled "Rising Waters - AI-Based Flood Prediction System". It features a navigation bar with links for Home, Predict, About, and Contact. The main content area is divided into two sections: "Enter Weather Parameters" and "Prediction Result".

Enter Weather Parameters:

- Rainfall (mm): 185
- Temperature (°C): 27
- Humidity (%): 89
- River Level (m): 8.5
- Wind Speed (km/h): 18

A blue button labeled "Predict Flood Risk" is located below the input fields. A note below the button states: "Please enter accurate values for better prediction."

Prediction Result:

- High Flood Risk:** There is a high chance of flood occurrence.
- Confidence Score:** 96.55%
- Recommended Action:** Evacuate low-lying areas and follow emergency guidelines. Stay updated with local weather reports and alerts.
- Input Summary:**
 - Rainfall: 185 mm
 - Temperature: 27 °C
 - Humidity: 89 %
 - River Level: 8.5 m
 - Wind Speed: 18 km/h

The footer of the application states: "© 2026 Rising Waters - AI-Based Flood Prediction System | All Rights Reserved".

Browser showing successful flood prediction:

The screenshot shows a web browser window with the URL `127.0.0.1:5000/result`. The application is titled "Rising Waters - AI-Based Flood Prediction System". It features a navigation bar with links for Home, Predict, About, and Contact. The main content area displays a success message: "Prediction Completed Successfully! Here is the flood risk prediction based on the provided weather parameters."

Flood Risk Level:

- High Flood Risk:** There is a high chance of flood occurrence in your area. Please take necessary precautions and stay updated.
- Confidence Score:** 96.55%

Input Parameters:

Parameter	Value
Rainfall	185 mm
Temperature	27 °C
Humidity	89 %
River Level	8.5 m
Wind Speed	18 km/h

Recommended Actions:

- Evacuate low-lying areas.
- Avoid unnecessary travel.
- Stay updated with local weather reports and alerts.
- Follow instructions from local authorities.

At the bottom of the main content area, there is an illustration of a house being flooded by waves, with a warning sign and a cloud raining. Below the illustration are two buttons: "Back to Home" and "Make Another Prediction".

The footer of the application states: "© 2026 Rising Waters - AI-Based Flood Prediction System | All Rights Reserved".